



Modelling of vortex-induced force and prediction of vortex-induced vibration of a bridge deck using method of multiple scales

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ABSTRACT

With the increase of span length, long-span bridges become more flexible and susceptible to vortex-induced vibrations (VIV). Several models for vortex-induced force (VIF) and various methods for predicting VIV have been developed accordingly but without consensus. This paper uses the method of multiple scales, from a different angle of view, to analyze the general polynomial model for VIF and present a concise polynomial model for VIF and a simple method for predicting VIV of a bridge deck. The aeroelastic vibration of a deck section is simulated by the computational fluid dynamics (CFD) and the results are used to verify the concise polynomial model of VIF. The forced vibration of a deck section is also simulated by CFD to yield the simple method for predicting VIV together with the frequency response equation and the critical equation of VIV derived by the method of multiple scales. The results show that the even and cross terms of aerodynamic damping and stiffness in the general polynomial VIF model are small compared with the odd terms and can be thus omitted. The lock-in region and the amplitude of VIV of the deck section can be accurately predicted by the proposed simple method.

1. Introduction

With the increase of span length, long-span bridges become more flexible and susceptible to vortex-induced vibrations (VIV). VIV refers to the phenomenon of wind-induced vibration in which the airflow passing through a bridge generates flow separation and periodic shedding of vortices, resulting in a periodic excitation on the bridge. In the initial stage of VIV, the external excitation dominates the vibration response. As time progresses, the vibration gradually stabilizes, and the self-excitation becomes the dominant factor. Therefore, VIV can be described as a limit cycle motion with both forced and self-excited characteristics (Bearman, 1984; Williamson and Govardhan, 2004). Although VIV does not directly cause a bridge to collapse, it can induce fatigue damage to crucial structural components of the bridge and impair the driving safety of vehicles (Fujino and Yoshida, 2002; Marra et al., 2011; Xin et al., 2018; Zhang et al., 2023). Several long-span bridges once suffered VIV, which include the Deer Isle Bridge in USA (Kumarasena et al., 1991), the Great Belt Bridge in Denmark (Larsen et al., 2000), the Second Severn Crossing in UK (Macdonald et al., 2002), the Yi Sun-Shin Bridge in Korea (Hwang et al., 2020), the Xihoumen Bridge and the Humen Bridge in China (Li et al., 2014; Ge et al., 2022).

Since VIV of long-span bridges often occurs at low wind speeds and may cause fatigue damage to the bridges, the accurate prediction of VIV of the bridge becomes important and necessary to make sure that the safety and serviceability of the bridge are not seriously affected by VIV.

The currently available prediction methods of VIV are mainly based on wind tunnel tests of the aeroelastic model of bridge deck section. These methods generally fall into two categories. The first category directly uses the results from sectional model tests to predict the maximum vortex-induced response (VIR) of the prototype bridge. In this category, a semi-empirical model of vortex-induced force (VIF) is first assumed. The maximum amplitude of VIR of the deck section, as a single degree of freedom (SDOF) system and by assuming that the VIV is harmonic in the lock-in region, is then derived from the semi-empirical VIF model as the function of several parameters. These parameters are then identified from wind-tunnel test results, and the maximum amplitude of VIR is finally estimated (Simiu and Scanlan, 1978, 1986; Zhu et al., 2017a). The second category uses wind tunnel test results to identify the parameters involved in semi-empirical models of VIF to obtain VIF time-histories and then applies the identified VIF to the analytical or numerical model of the bridge for VIR prediction (Zhu et al., 2018). Barhoush et al. (1995) used a 2DOF semi-empirical VIF model and a

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finite element (FE) beam model to obtain the solution of VIR. In this 2DOF semi-empirical VIF model, the structural motion is modeled by a second-order linear mechanical oscillator excited by the VIF modeled by a second-order nonlinear wake oscillator (Vol del Pol type oscillator) coupled with the structural motion (Bishop and Hassan, 1964a,b; Hartlen and Currie, 1970). By using a different 2DOF semi-empirical VIF model and a FE bridge model, Diana et al. (2006) solved the VIV problem. Nevertheless, the experimental procedures involved to identify the parameters in the 2DOF VIF model are complicated with considerable uncertainty (Wu and Kareem, 2012). Moreover, Ehsan and Scanlan (1990) modified Scanlan's linear SDOF VIV model by incorporating nonlinear aerodynamic damping to form a nonlinear SDOF VIV model and proposed an experimental procedure to identify the parameters in the nonlinear SDOF VIF model. Goswami et al. (1993) improved the Scanlan's empirical nonlinear model by considering the forced effect and introducing nonlinear damping into the parameterized vibration equation. Mashnad and Jones (2014) extended the aerodynamic damping and stiffness terms of the model to the nonlinear functions of structural amplitude. Hansen (2013) introduced turbulence intensity correction based on the Vickery-Basu spectrum method and applied it to simulate vortex-induced vibration of bridge sections. In consideration that the VIV of a bridge is often excited by one mode at a time, Meng et al. (2015) proposed a mode-by-mode solution method for VIV of a prototype bridge based on a different nonlinear SDOF VIF model proposed by Zhu et al. (2013). Bishop and Hassan (1964a,b) proposed a two-degree-of-freedom vortex-induced force model, which approximates the fluid-structure coupling effect by adding a coupled nonlinear oscillator equation (e.g., Van Der Pol oscillator) to the structural motion equation. Subsequent researchers have conducted extensive studies on this type of model, focusing on the selection of nonlinear equations (Gabbai and Benaroya, 2005), the form of fluid-structure coupling (Facchinetti et al., 2004), improvement of model parameters (Ogink and Metrikine, 2010), and extension from two-dimensional to three-dimensional models (Srinil et al., 2009). More information on the existing models and experiments performed on the VIV of bridge decks can be found in the review articles by Ge et al. (2019); Zhu et al. (2010).

With the rapid growth in computational capacity, the computational fluid dynamics (CFD) is becoming a powerful tool for simulating VIV of a deck section. Sun et al. (2009) examined the suitability of different turbulence models for simulating aeroelastic problems in the Reynolds Averaged Navier-Stokes (RANS) framework and concluded that $k-\omega$ model is an appropriate choice balancing the computational efficiency and accuracy. Fan et al. (2022) conducted numerical simulations using Computational Fluid Dynamics (CFD) to analyze the aerodynamic interference between upper and lower box girders. They discussed the variations in aerodynamic characteristics with different angles of attack and distances between the two girders, and investigated the influence of the wind environment on the vehicle's aerodynamic coefficients. To avoid large computational cost involved in aeroelastic vibration simulation, Noguchi et al. (2020) predicted the steady-state VIV amplitude of a box girder deck section using forced vibration simulation with coordinate transformation in large-eddy simulation (LES). However, the predicted vibration amplitude was much larger than that from aeroelastic wind tunnel tests. Wang and Chen (2022) also adopted forced vibration simulation but used high-order spectral element/Fourier method with coordinate transformation to extract the aerodynamic damping induced by VIV and predict VIV of the deck section. A good match of VIV amplitudes between simulations and wind tunnel tests was achieved. Nevertheless, the above aeroelastic and forced vibration simulations of VIV of the deck section are limited to 2D domain and there were no discussions on the VIF models and the conversion of the simulated VIV results to the prototype bridge.

From a different angle of view, this paper uses the method of multiple scales to analyze the general polynomial model for VIF and to present a concise model for VIF and a simple method for predicting VIV of a deck section. In fact, the general polynomial model for VIF covers all the

above-mentioned linear and nonlinear SDOF VIF models. The aeroelastic vibration simulation of a deck section by CFD is used to verify the proposed concise model for VIF. The forced vibration simulation of a deck section by CFD is used, together with the frequency response equation and critical equation of VIV derived by the method of multiple scales, to provide the simple method for predicting VIV of the deck section. The remaining parts of this paper are organized as follows. Section 2 introduces the modelling of VIF and the derivation of frequency response equation and critical equation of VIV using the method of multiple scales. Section 3 presents the aeroelastic vibration simulation of a deck section and the verification of the proposed concise VIF model. Section 4 provides the proposed simple method for predicting VIV and the forced vibration simulation of a deck section. Section 5 shows the results of a case study for predicting VIV. The concluding remarks on this study are given in Section 6.

2. Modelling of VIF by method of multiple scales

2.1. General polynomial model for VIF

In most of the existing SDOF VIF models, VIF is expressed as a function of displacement and velocity of a deck section in a polynomial form. The Scanlan's linear and nonlinear vortex-induced force models for a SDOF system of the deck section in vertical vibration are expressed by Eq. (1) and Eq. (2), respectively (Simiu and Scanlan, 1996).

$$F_{VI} = \frac{1}{2} \rho U^2 D \left[Y_1(K) \frac{\dot{x}}{U} + Y_2(K) \frac{x}{D} + C_L(K) \sin(\omega_s t + \varphi) \right] \quad (1)$$

$$F_{VI} = \frac{1}{2} \rho U^2 D \left[Y_1(K) \left(1 - \varepsilon_1 \frac{x^2}{D^2} \right) \frac{\dot{x}}{U} + Y_2(K) \frac{x}{D} + C_L(K) \sin(\omega_s t + \varphi) \right] \quad (2)$$

where ρ is the air density; U is the incoming wind speed; D is the characteristic dimension and often refers to the crosswind width of the deck section; $Y_1(K)$, $Y_2(K)$, $C_L(K)$ and ε_1 are the aerodynamic parameters of VIF, which could be identified using either wind tunnel tests or CFD simulations; K is the reduced frequency of vortex shedding and $K = \omega_s D / U$; ω_s is the circle frequency of the VIF; t is the time; φ is the phase difference; x , $\dot{x}$ is the displacement and velocity of deck section vibration in vertical direction.

Using directly-measured VIF time-histories from a deck section in vertical direction in a wind tunnel, Zhu et al. (2013) found that although the Scanlan's models are useful for predicting the maximum amplitude of VIR, but it does not fit the exact time-history of the measured VIF. Zhu et al. (2013) then proposed a semi-empirical VIF model for a single-box deck section as shown in Eq. (3).

$$F_{VI} = \frac{1}{2} \rho U^2 D \left[Y_1(K) \left(1 + \varepsilon_{03} \frac{x^2}{U^2} + \varepsilon_{11} \frac{x}{D} \right) \frac{\dot{x}}{U} + Y_2(K) \frac{x}{D} + C_L(K) \sin(\omega_s t + \varphi) \right] \quad (3)$$

Eq. (3) was proposed based on the wind tunnel tests of a single-box deck section. Later, because the terms used in the VIF model expressed by Eq. (3) are not sufficient to reproduce the measured VIF time-histories for a twin-box deck section and a semi-closed deck section, Zhu et al. (2017b) further proposed a semi-empirical VIF model for a twin-box deck section and a semi-closed deck section, expressed by Eq. (4) and Eq. (5) respectively.

$$F_{VI} = \frac{1}{2} \rho U^2 D \left[Y_1(K) \left(1 + \varepsilon_{03} \frac{x^2}{U^2} + \varepsilon_{04} \frac{\dot{x}^3}{U^3} + \varepsilon_{11} \frac{x}{D} + \varepsilon_{22} \frac{x^2}{D^2} \frac{\dot{x}}{U} \right) \frac{\dot{x}}{U} + Y_2(K) \left(1 + \zeta_2 \frac{x}{D} \right) \frac{x}{D} + C_L(K) \sin(\omega_s t + \varphi) \right] \quad (4)$$

$$F_{VI} = \frac{1}{2} \rho U^2 D \left[Y_1(K) \left(1 + \varepsilon_{03} \frac{\dot{x}^2}{U^2} + \varepsilon_{11} \frac{x}{D} \right) \frac{\dot{x}}{U} + Y_2(K) \left(1 + \zeta_2 \frac{x}{D} \right) \frac{x}{D} + C_L(K) \sin(\omega_s t + \varphi) \right] \quad (5)$$

In Eq. (3)-Eq. (5), ε_{03} , ε_{04} , ε_{11} , ε_{22} , and ζ_2 are all the aerodynamic parameters to be identified using either wind tunnel tests or CFD simulations.

By observation of all the above VIF models, it can be inferred that the most general form of polynomial VIF models can be expressed by Eq. (6).

$$F_{VI} = \frac{1}{2} \rho U^2 D \left[\sum_{i=1}^n p_i \left(\frac{\dot{x}}{U} \right)^i + \sum_{j=1}^k q_j \left(\frac{x}{D} \right)^j + \sum_{i,j=1}^{n,k} r_{ij} \left(\frac{\dot{x}}{U} \right)^i \left(\frac{x}{D} \right)^j + C_L(K) \sin(\omega_s t + \varphi) \right] \quad (6)$$

where p_i , q_j and r_{ij} ($i = 1, 2, \dots, n; j = 1, 2, \dots, k$) are all the aerodynamic parameters of VIF.

2.2. Concise polynomial model for VIF and method of multiple scales

Based on Eq. (6), the governing equation of a deck section modeled as a SDOF under vertical VIF can be written as

$$m\ddot{x} + 2m\omega_0 \xi_m \dot{x} + m\omega_0^2 x = \frac{1}{2} \rho U^2 D \left[\sum_{i=1}^n p_i \left(\frac{\dot{x}}{U} \right)^i + \sum_{j=1}^k q_j \left(\frac{x}{D} \right)^j + \sum_{i,j=1}^{n,k} r_{ij} \left(\frac{\dot{x}}{U} \right)^i \left(\frac{x}{D} \right)^j + C_L(K) \sin(\omega_s t + \varphi) \right] \quad (7)$$

where m is the mass per unit length of the deck section; ξ_m is the me-

$$f(x, \dot{x}) = \frac{1}{2} \rho U^2 D \sum_{i=1}^n p_i \left(\frac{\dot{x}}{U} \right)^i - 2m\omega_0 \xi_m \dot{x} + \frac{1}{2} \rho U^2 D \sum_{j=1}^k q_j \left(\frac{x}{D} \right)^j + \frac{1}{2} \rho U^2 D \cdot \sum_{i,j=1}^{n,k} r_{ij} \left(\frac{\dot{x}}{U} \right)^i \left(\frac{x}{D} \right)^j + \frac{1}{2} \rho U^2 D C_L(K) \sin(\omega_s t + \varphi) \quad (9)$$

According to the method of multiple scales, the time scales $T_0, T_1, \dots, T_n$ are introduced as

$$T_n = \varepsilon^n t \quad (n = 0, 1, 2, \dots) \\ T_0 = t, T_1 = \varepsilon t, T_2 = \varepsilon^2 t, \dots \quad (10)$$

Since the $n+1$ times scales are introduced, the displacement response of the deck section, $x(t; \varepsilon)$, is equal to $x(T_0, T_1, T_2, \dots, T_n; \varepsilon)$, and the solution of Eq. (8) can be expressed as

$$x(t; \varepsilon) = \sum_{i=0}^n \varepsilon^i x_n(T_0, T_1, T_2, \dots, T_n) + o(\varepsilon^{n+1}) \quad (11)$$

where $o(\varepsilon^{n+1})$ is the error for the finite series of $x(t; \varepsilon)$. According to the time scale and the differential rule of multiple functions, the time derivative can be expressed by Eq. (12)

$$\dot{x} = \frac{dx}{dt} = \frac{dT_0}{dt} \frac{\partial x}{\partial T_0} + \frac{dT_1}{dt} \frac{\partial x}{\partial T_1} + \frac{dT_2}{dt} \frac{\partial x}{\partial T_2} + \dots = D_0 x + \varepsilon D_1 x + \varepsilon^2 D_2 x + \dots \\ \ddot{x} = \frac{d\dot{x}}{dt} = \frac{dT_0}{dt} \frac{\partial \dot{x}}{\partial T_0} + \frac{dT_1}{dt} \frac{\partial \dot{x}}{\partial T_1} + \frac{dT_2}{dt} \frac{\partial \dot{x}}{\partial T_2} + \dots = D_0 \dot{x} + \varepsilon D_1 \dot{x} + \varepsilon^2 D_2 \dot{x} + \dots \quad (12)$$

where $D_0, D_1, D_2 \dots$ are the symbols for taking the partial derivatives of $T_0, T_1, T_2 \dots$

The substitution of Eq. (11) into Eq. (12) yields

$$\dot{x} = D_0(x_0 + \varepsilon x_1 + \dots) + \varepsilon D_1(x_0 + \varepsilon x_1 + \dots) + \dots = D_0 x_0 + \varepsilon D_0 x_1 + \varepsilon D_1 x_0 + \varepsilon^2 D_1 x_1 + \dots \\ \ddot{x} = D_0(D_0 x_0 + \varepsilon D_0 x_1 + \varepsilon D_1 x_0 + \varepsilon^2 D_1 x_1 + \dots) + \varepsilon D_1(D_0 x_0 + \varepsilon D_0 x_1 + \varepsilon D_1 x_0 + \varepsilon^2 D_1 x_1 + \dots) + \dots = D_0^2 x_0 + \varepsilon D_0^2 x_1 + 2\varepsilon D_0 D_1 x_0 + 2\varepsilon^2 D_0 D_1 x_1 + \dots \quad (13)$$

chanical damping ratio of the deck section; ω_0 is the natural circle frequency of the deck section; $x, \dot{x}, \ddot{x}$ is the displacement, velocity and acceleration of the deck section vibration in vertical direction.

By observation of the VIV results from either field measurement, numerical simulation or wind tunnel test, it can be found that the displacement time histories of VIV of the deck section are quasi-periodic (Fujino and Yoshida, 2002; Li et al., 2011; Xin et al., 2012; Qin et al., 2014). It is sometime even assumed that vortex-induced displacement response is harmonic when predicting the maximum amplitude of VIV of a deck section (Simiu and Scanlan, 1978, 1986; Zhu et al., 2017a). Chen (2012) also pointed out that VIV is a weakly nonlinear oscillator and then derived the approximate solution for VIV in the lock-in region. In this study, the VIV of a deck section as a SDOF is also regarded as a weakly nonlinear oscillator, and the method of multiple scales (Sanchez, 1996) is then used to analyze the general polynomial VIF model and to find solution of Eq. (7). In this regard and considering that the mechanical damping ratio is a small quantity, Eq. (7) can be re-written as Eq. (8).

$$m\ddot{x} + m\omega_0^2 x = \varepsilon f(x, \dot{x}) \quad (8)$$

where $0 < \varepsilon \ll 1$ and $f(x, \dot{x})$ is as follows:

The field measurement, numerical simulation and wind tunnel test all shows that in the lock-in region of VIV, the vortex shedding frequency is approximately equal to the natural frequency of the deck section as a SDOF (Fujino and Yoshida, 2002; Belloli et al., 2014; Yang et al., 2021; Dong et al., 2022) but the two frequencies are not exactly same. Therefore, the tuning parameter σ as a small quantity is introduced to represent the difference between the vortex shedding frequency and the natural frequency of the deck section.

$$\varepsilon \sigma = \omega_s - \omega_0 \quad (14)$$

$$\omega_s t + \varphi = \omega_0 t + \varepsilon \sigma t + \varphi = \omega_0 T_0 + \sigma T_1 + \varphi \quad (15)$$

Substituting Eq. (11) - Eq. (15) into Eq. (8) and sorting all the terms according to the same power of ε yield

$$\varepsilon^0 \cdot (m D_0^2 x_0 + m \omega_0^2 x_0) + \varepsilon^1 \cdot [m D_0^2 x_1 + m \omega_0^2 x_1 + 2m D_0 D_1 x_0 + 2m \omega_0 \xi_m D_0 D_1 x_0 - \frac{1}{2} \rho U^2 D \sum_{i=1}^n p_i \left(\frac{D_0 x_0}{U} \right)^i - \frac{1}{2} \rho U^2 D \sum_{j=1}^k q_j \left(\frac{x_0}{D} \right)^j - \frac{1}{2} \rho U^2 D \sum_{i,j=1}^{n,k} r_{ij} \left(\frac{D_0 x_0}{U} \right)^i \left(\frac{x_0}{D} \right)^j - \frac{1}{2} \rho U^2 D C_L(K) \sin(\omega_0 T_0 + \sigma T_1 + \varphi)] + o(\varepsilon^1) = 0 \quad (16)$$

If there is a solution for Eq. (16), the following equations should exist:

$$\begin{cases} mD_0^2x_0 + m\omega_0^2x_0 = 0 \\ mD_0^2x_1 + m\omega_0^2x_1 = -2mD_0D_1x_0 - 2m\omega_0\xi_m D_0x_0 + \frac{1}{2}\rho U^2 D \sum_{i=1}^n p_i \left(\frac{D_0x_0}{U}\right)^i + \frac{1}{2}\rho U^2 D \cdot \sum_{j=1}^k q_j \left(\frac{x_0}{D}\right)^j + \frac{1}{2}\rho U^2 D \sum_{i=1}^{n,k} r_{ij} \left(\frac{D_0x_0}{U}\right)^i \left(\frac{x_0}{D}\right)^j \\ \quad + \frac{1}{2}\rho U^2 DC_L(K) \sin(\omega_0 T_0 + \sigma T_1 + \varphi) \\ \dots \end{cases} \quad (17)$$

Each differential equation is linear and can be solved sequentially. The solution of Eq. (7) is therefore approximated in the form of gradual approximation. That is, the more the number of terms in Eq. (17) is taken, the more the degree of approximation.

The system simplified by removing the small terms associated with the higher powers of ε is called a derived system, and any periodic solution of the derived system is called a derived solution. In this study, the derived system and derived solution of Eq. (7) are sought by considering the solutions of the first two differential equations, Eq.(17a) and Eq. (17b). The derived solution of Eq.(17a) is

$$x_0 = a \cos (\omega_0 T_0 + \theta) \quad (18)$$

where a and θ are the undetermined real functions of T_1 . The substitution of Eq. (18) into the second differential equation, Eq.(17b), leads to

$$\begin{aligned} mD_0^2x_1 + m\omega_0^2x_1 &= A_0 \sin(\omega_0 T_0 + \theta) + B_0 \cos (\omega_0 T_0 + \theta) + C_0 + D_0 + E_0 + \dots \\ A_0 &= 2m\omega_0 D_1 a + 2m\omega_0^2 \xi_m a - \frac{1}{2}\rho U^2 D \sum_{i=1}^n p_{2i-1} \left(\frac{\omega_0 a}{U}\right)^{2i-1} + \frac{1}{2}\rho U^2 DC_L(K) \cos(\sigma T_1 + \varphi - \theta) \\ B_0 &= 2m\omega_0 a D_1 \varphi + \frac{1}{2}\rho U^2 D \sum_{j=1}^k q_{2j-1} \left(\frac{a}{D}\right)^{2j-1} - \frac{1}{2}\rho U^2 DC_L(K) \sin(\sigma T_1 + \varphi - \theta) \\ C_0 &= \frac{1}{2}\rho U^2 D \sum_{i=1}^n p_{2i} \left(\frac{\omega_0 a}{U}\right)^{2i} \sin^2(\omega_0 T_0 + \theta) \\ D_0 &= \frac{1}{2}\rho U^2 D \sum_{j=1}^k q_{2j} \left(\frac{a}{D}\right)^{2j} \cos^2(\omega_0 T_0 + \theta) \\ E_0 &= \frac{1}{2}\rho U^2 D \sum_{i=1}^{n,k} r_{ij} \frac{a^{i+j} \omega_0^i}{U^i D^j} \cos^i(\omega_0 T_0 + \theta) \sin^j(\omega_0 T_0 + \theta) \end{aligned} \quad (19)$$

$$\begin{aligned} mD_0^2x_1 + m\omega_0^2x_1 &= A_0 \sin(\omega_0 T_0 + \theta) + B_0 \cos (\omega_0 T_0 + \theta) \\ A_0 &= 2m\omega_0 D_1 a + 2m\omega_0^2 \xi_m a - \frac{1}{2}\rho U^2 D \sum_{i=1}^n p_{2i-1} \left(\frac{\omega_0 a}{U}\right)^{2i-1} + \frac{1}{2}\rho U^2 DC_L(K) \cos(\sigma T_1 + \varphi - \theta) \\ B_0 &= 2m\omega_0 a D_1 \varphi + \frac{1}{2}\rho U^2 D \sum_{j=1}^k q_{2j-1} \left(\frac{a}{D}\right)^{2j-1} - \frac{1}{2}\rho U^2 DC_L(K) \sin(\sigma T_1 + \varphi - \theta) \end{aligned} \quad (20)$$

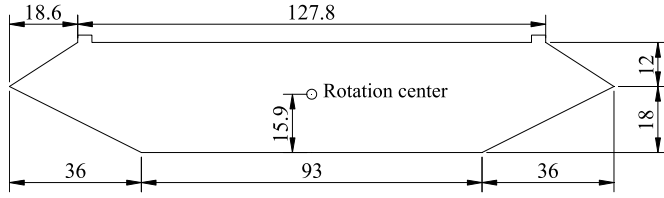


Fig. 1. Configuration of deck section (unit:mm).

Correspondingly, the concise polynomial VIF model can be obtained as.

$$F_{VI} = \frac{1}{2} \rho U^2 D \left[\sum_{i=1}^n p_{2i-1} \left(\frac{\dot{x}}{U} \right)^{2i-1} + \sum_{j=1}^k q_{2j-1} \left(\frac{x}{D} \right)^{2j-1} + C_L(K) \sin(\omega_s t + \varphi) \right] \quad (21)$$

Compared with the general polynomial VIF model, Eq. (6), the even and cross terms of aerodynamic damping and stiffness are neglected in the concise polynomial VIF model expressed by Eq. (21). Since this study focuses on the vortex-induced aerodynamic force, which is listed on the right side of the dynamic equation of the bridge deck, the plus or minus values of the aerodynamic damping force terms has the opposite physical meaning compared with the mechanical damping force: plus value (not minus value) means excitation of the oscillation.

2.3. Frequency response equation and critical equation

In the context of the multiscale method, periodic solutions play a crucial role in describing the behavior of the system that exhibits a

significant difference in time scales. To accurately capture the dynamic behavior of such a system, it is essential to ensure that the average value of the periodic solution remains zero. It is therefore necessary to set the pre-factors of the sine and cosine functions in Eq. (20) to zero, i.e., $A_0 = 0$ and $B_0 = 0$, to prevent the introduction of any system bias (Karimiasl et al., 2019). As such, the following equation, i.e., Eq. (22), can be derived.

$$\begin{cases} D_1 a = -\omega_0 \xi_m a + \frac{\rho U^2 D \sum_{i=1}^n p_{2i-1} \left(\frac{\omega_0 a}{U} \right)^{2i-1}}{4m\omega_0} - \frac{\rho U^2 D C_L(K)}{4m\omega_0} \cos(\sigma T_1 + \varphi - \theta) \\ a D_1 \theta = -\frac{\rho U^2 D \sum_{j=1}^k q_{2j-1} \left(\frac{a}{D} \right)^{2j-1}}{4m\omega_0} + \frac{\rho U^2 D C_L(K)}{4m\omega_0} \sin(\sigma T_1 + \varphi - \theta) \end{cases} \quad (22)$$

Eq. (22) can be converted to the form of a self-consistent system without explicit T_1 . To this end, $\beta = \sigma T_1 + \varphi - \theta$ is introduced to Eq. (22), and one can obtain:

$$\begin{cases} D_1 a = -\omega_0 \xi_m a + \frac{\rho U^2 D \sum_{i=1}^n p_{2i-1} \left(\frac{\omega_0 a}{U} \right)^{2i-1}}{4m\omega_0} - \frac{\rho U^2 D C_L(K)}{4m\omega_0} \cos \beta \\ a D_1 \beta = a \sigma + \frac{\rho U^2 D \sum_{j=1}^k q_{2j-1} \left(\frac{a}{D} \right)^{2j-1}}{4m\omega_0} - \frac{\rho U^2 D C_L(K)}{4m\omega_0} \sin \beta \end{cases} \quad (23)$$

Let $D_1 a = D_1 \beta = 0$ to obtain the equations of determining the singularity (a, β).

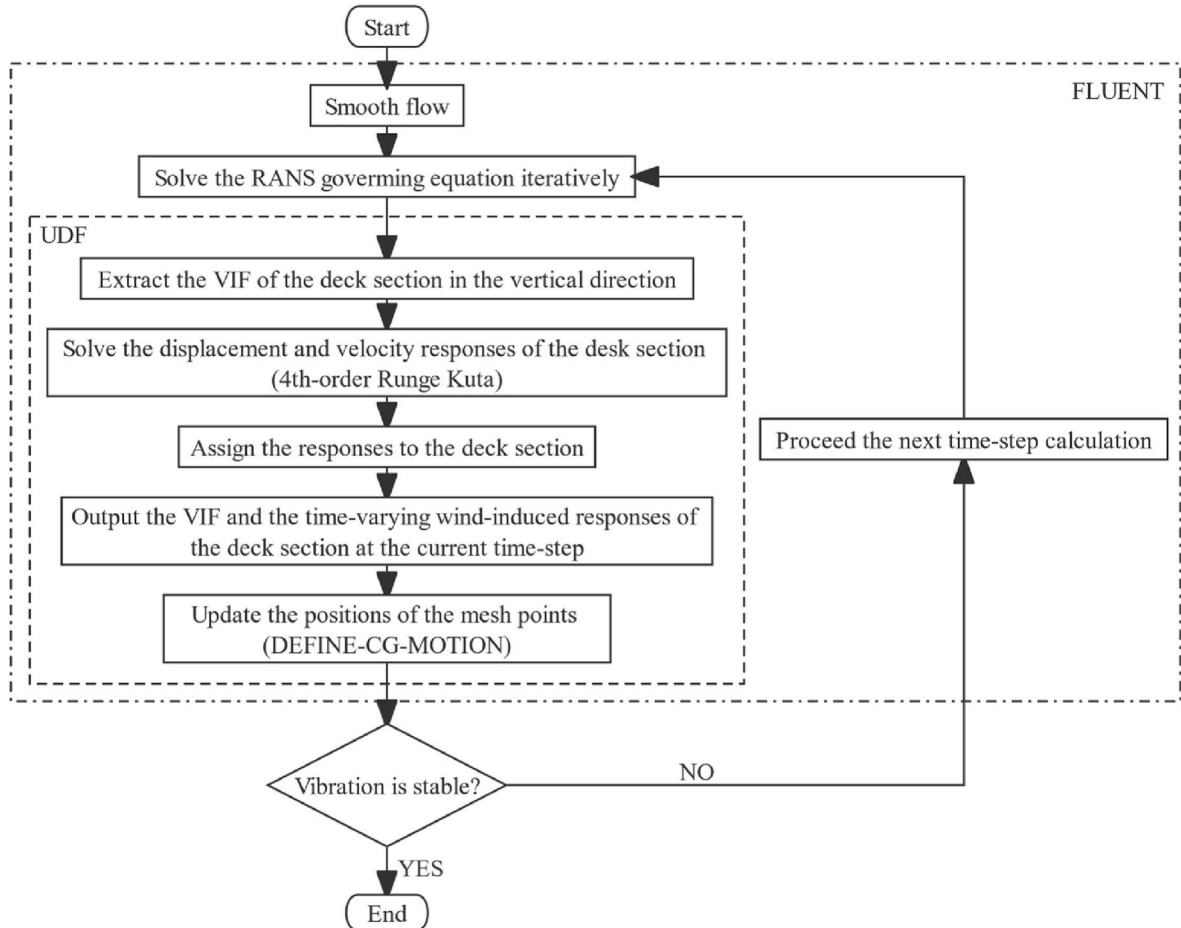


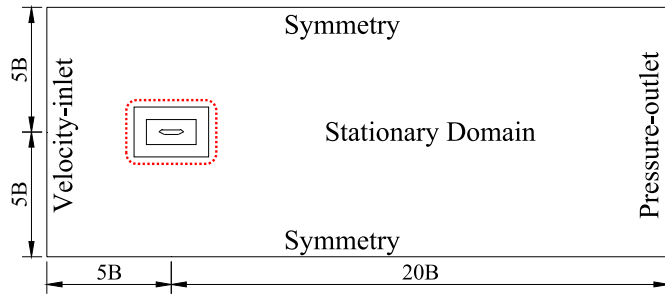
Fig. 2. Flow chart of aeroelastic vibration simulation.

$$\begin{cases} \omega_0 \xi_m a - \frac{\rho U^2 D \sum_{i=1}^n p_{2i-1} \left(\frac{\omega_0 a}{U} \right)^{2i-1}}{4m\omega_0} = \frac{\rho U^2 D C_L(K)}{4m\omega_0} \cos \beta \\ a\sigma + \frac{\rho U^2 D \sum_{j=1}^k q_{2j-1} \left(\frac{a}{D} \right)^{2j-1}}{4m\omega_0} = \frac{\rho U^2 D C_L(K)}{4m\omega_0} \sin \beta \end{cases} \quad (24)$$

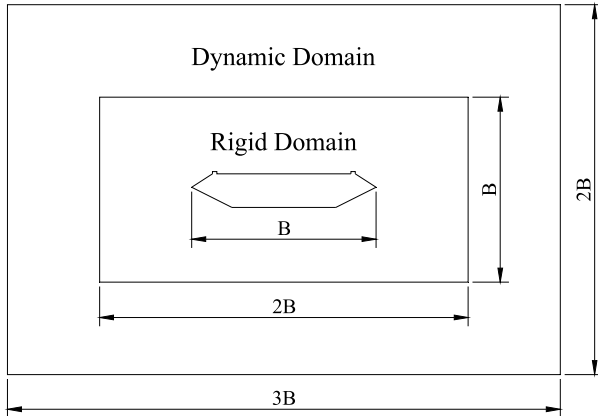
By squaring the two terms of Eq. (24) and summing them, one can obtain

$$\begin{aligned} \omega_0^2 a^2 \left(2m\omega_0 \xi_m - \frac{1}{2} \rho U^2 D \sum_{i=1}^n \frac{p_{2i-1} \omega_0^{2i-2} a^{2i-2}}{U^{2i-1}} \right)^2 + \left(2m\omega_0 a \sigma + \frac{1}{2} \rho U^2 D \sum_{j=1}^k \frac{q_{2j-1} a^{2j-1}}{D^{2j-1}} \right)^2 \\ = \left(\frac{1}{2} \rho U^2 D C_L(K) \right)^2 \end{aligned} \quad (25)$$

$$\begin{aligned} \omega_0^2 a^2 \left(2m\omega_0 \xi_m - \frac{1}{2} \rho U^2 D \sum_{i=1}^n \frac{p_{2i-1} \omega_0^{2i-2} a^{2i-2}}{U^{2i-1}} \right) \cdot \left(2m\omega_0 \xi_m - \rho U^2 D \sum_{i=1}^n \left(\frac{2i-1}{2} \right) \frac{p_{2i-1} \omega_0^{2i-2} a^{2i-2}}{U^{2i-1}} \right) \\ + \left(2m\omega_0 a \sigma + \frac{1}{2} \rho U^2 D \sum_{j=1}^k \frac{q_{2j-1} a^{2j-2}}{D^{2j-1}} \right) \cdot \left(2m\omega_0 a \sigma + \rho U^2 D \sum_{j=1}^k \left(\frac{2j-1}{2} \right) \frac{q_{2j-1} a^{2j-2}}{D^{2j-1}} \right) = 0 \end{aligned} \quad (26)$$



(a) overall



(b) local

Fig. 3. Computation domain and boundary conditions of deck section.

Eq. (25) can be used to determine the response amplitude a as a function of the natural frequency ω_0 or the vortex shedding frequency ω_s . Therefore, Eq. (25) is called as the frequency response equation. Furthermore, “ $da/d\sigma = 0$ ” can be obtained from the frequency response equation, that is, the trajectory equation of a point with a straight tangent line. This equation is the boundary line between the stability and instability of VIV, and it is called the critical equation.

Based on the frequency response equation and the critical equation, a simple method is proposed to predict VIR of a deck section in Section 4.

3. Aeroelastic vibration simulation and verification of concise VIF model

In this section, the aeroelastic vibration simulation of VIV of a deck section using CFD is performed to verify the concise VIF model, that is, to verify if the even and cross terms of aerodynamic damping and stiffness can be neglected compared with the odd terms.

3.1. Aeroelastic vibration simulation by CFD

The deck section used by [Noguchi et al. \(2020\)](#) for investigating VIV in smooth flow in a wind tunnel is selected for the present aeroelastic vibration simulation. The section size is shown in [Fig. 1](#). The width B is 165 mm, the height D is 30 mm, the side length ratio B/D is 5.5, the mass per unit length of the deck section is 2.016 kg/m, and the natural frequency of the section in the vertical direction is 3.104Hz. In the aeroelastic vibration simulation, only the vertical vibration of the 2D deck section is considered with a mechanical damping ratio of 0.1%.

The commercial software ANSYS FLUENT is used to simulate the flow around and VIV acting on the deck section. The shear stress transport (SST) $k-\omega$ turbulence model in the form of two equations is adopted to relate the Reynolds stresses to the time-averaged variables. The finite volume method is used to discrete the Reynolds Averaged Navier-Stokes (RANS) governing equations in a second order format. The governing equations are given in Arbitrary Lagrange-Euler (ALE) formulations, which accommodate the moving boundaries and any subsequent deformation of the underlying discrete meshes. The gradient term is discretized based on the least square element method. The pressure term is discretized in the second-order format, and the momentum, turbulent kinetic energy and specific dissipation rate are

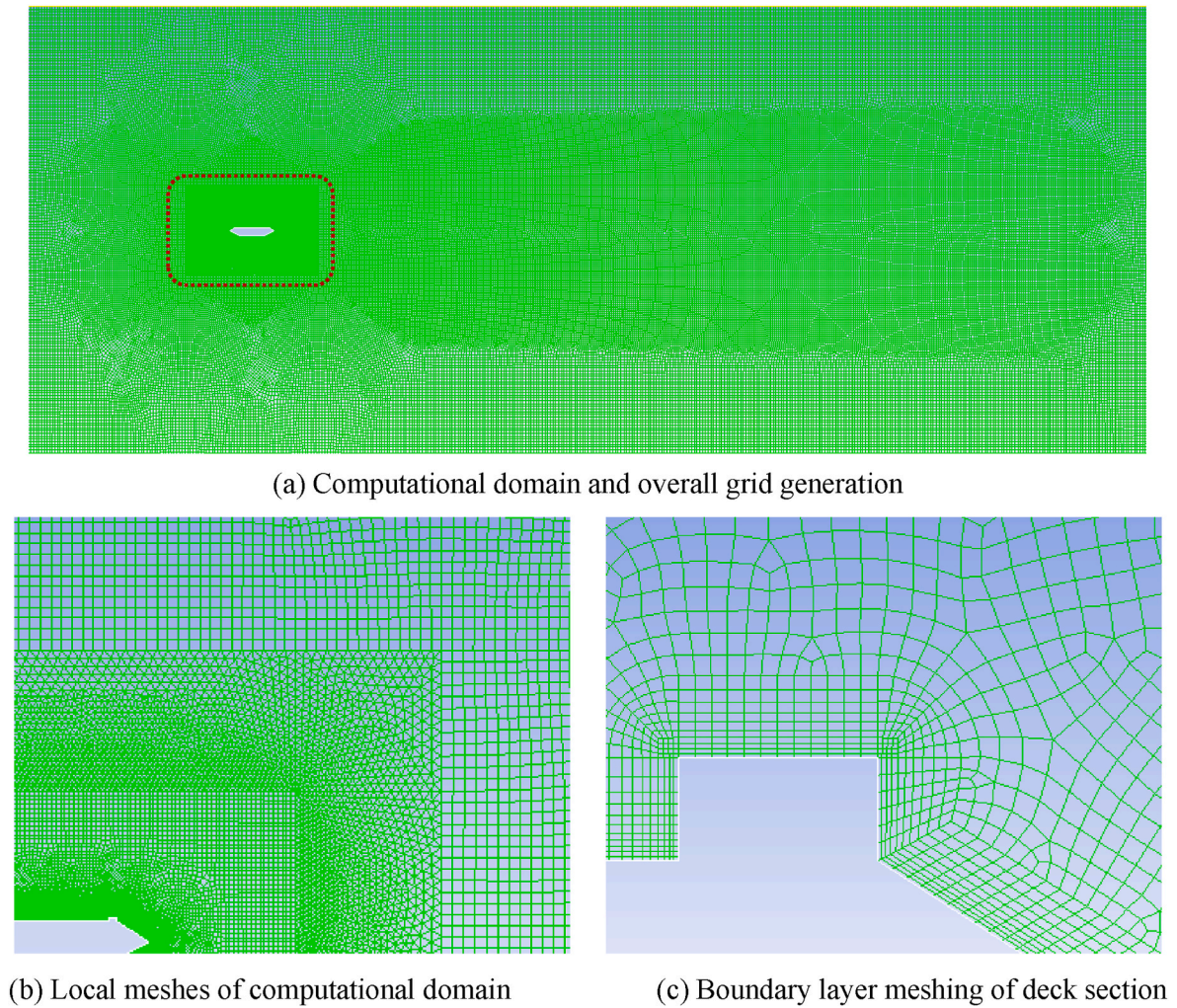


Fig. 4. Grid generation of deck section.

Table 1
Computed maximum displacement using different time-steps.

Time step (s)	Computed results (mm)	Test results (mm)	Error (%)
0.001005	3.96	3.91	1.28
0.00201	3.97		1.53
0.003015	3.88		−0.77
0.00402	3.81		−2.56
0.00603	3.75		−4.09
0.00804	3.68		−5.88

Table 2
Grid schemes and boundary layer settings.

Grid	First grid layer height (m)	y^+ max	Number of grid layers in the boundary	Grid growth rate	Grid number
G1	0.0000375	0.133	12	1.15	150233
G2	0.00005	0.188	12	1.14	145171
G3	0.00005	0.235	10	1.13	141178
G4	0.000075	0.283	10	1.12	137884
G5	0.0000875	0.331	10	1.10	134456
G6	0.00001	0.377	10	1.08	131424

Table 3
Computed maximum displacement under different mesh divisions.

Grid	Computed results (mm)	Test results (mm)	Error (%)
G1	3.88	3.91	−0.77
G2	3.88		−0.77
G3	3.88		−0.77
G4	3.89		−0.51
G5	3.85		−1.53
G6	3.83		−2.05

discretized in the QUICK format. The pressure-velocity coupling is performed using the SIMPLEC algorithm.

The weak coupling method is used to solve the fluid-structure interaction (FSI) problem of the present study (Ying et al., 2017). In the weak coupling method, the fluid and structural governing equations are sequentially solved. Therefore, the existing solvers for the fluid and structure can be easily adopted, and the flexibility between the fluid solver and structure solver is maintained. The computational process of the aeroelastic vibration simulation of the deck section is shown in Fig. 2. Firstly, the RANS governing equation for the deck section is solved iteratively under a given incoming wind speed, and the VIF of the deck section in the vertical direction is extracted using the fluid solver. Then, the VIF is applied to the deck section and the vortex-induced displacement and velocity responses of the deck section are solved using the 4th-order Runge Kutta method (the structure solver). The

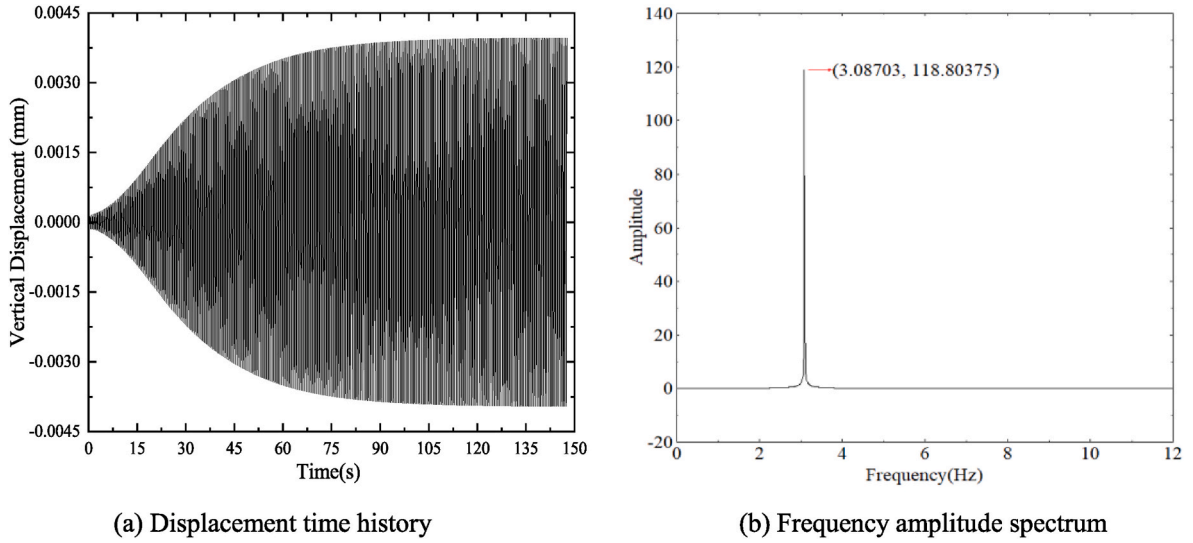


Fig. 5. Displacement time history and frequency amplitude spectrum of deck section.

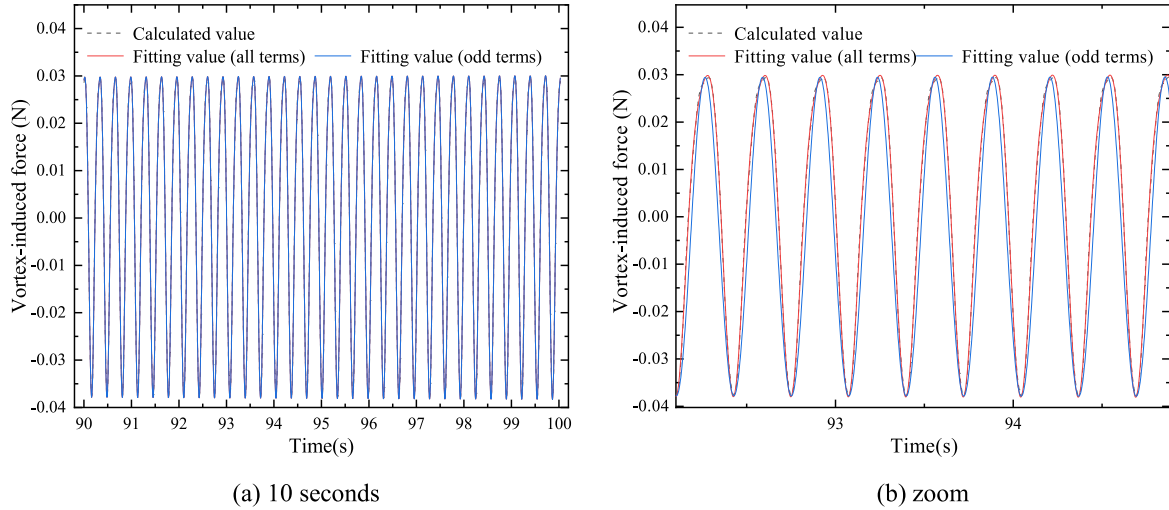


Fig. 6. Comparison of VIF time histories.

velocity response is then used to determine the new position for the deck section at the next time-step computation using the fluid solver. The positions of the mesh points are also calculated and updated using the spring-based smoothing technique. On the updated mesh, the instantaneous flow field around the deck section can be obtained using the unsteady fluid solver, and the corresponding aerodynamic forces can be calculated synchronously. Repeat the above steps, the time-varying wind-induced responses and aerodynamic forces can be determined step by step.

3.2. Computation domain, boundary condition and grid generation

The boundary conditions of aeroelastic vibration simulation are shown in Fig. 3a. The inlet boundary is a velocity inlet, and the outlet boundary is a pressure outlet. The symmetric boundary conditions are adopted at the upper and lower boundaries. The velocity inlet distanced from the centroid of the deck section is $5B$ (B is the width of deck section). The velocity outlet distanced from the centroid is $20B$. The upper and lower boundary distanced from the centroid is $5B$. The blockage ratio of the computation domain is 1.82%.

The computation domain and the grid generation of the deck section

are shown in Figs. 3b and 4 respectively. The rigid domain, dynamic domain and stationary domain are set in the computing domain (see Fig. 3b). The rigid domain is $2B \times B$ rectangle, which moves with the deck section, to ensure the simulation accuracy of the fluid around the bridge. The meshes with ten boundary layers are set on the section surface and the growth rate is 1.15. The deck section surface is therefore a non-sliding wall. Unstructured quadrilateral meshes are used for the remaining part. To meet the requirements of up and down motion of the rigid domain, a dynamic domain with the length and height of $3B \times 2B$ is set outside the rigid domain, and a triangular mesh with good elasticity smoothness and local reconstruction properties is set inside this domain. The remaining peripheral area of the computing domain is called the stationary domain, and the mesh type is unstructured quadrilateral meshes. To improve the computational efficiency, the outer region is discretized by a slightly larger grid. The overall mesh distortion rate is less than 3%.

3.3. Sensitivity study of time step and meshing size

Six time-steps of 0.001005s, 0.00201s, 0.003015s, 0.00402s, 0.00603s and 0.00804s are selected to carry out a sensitivity study of

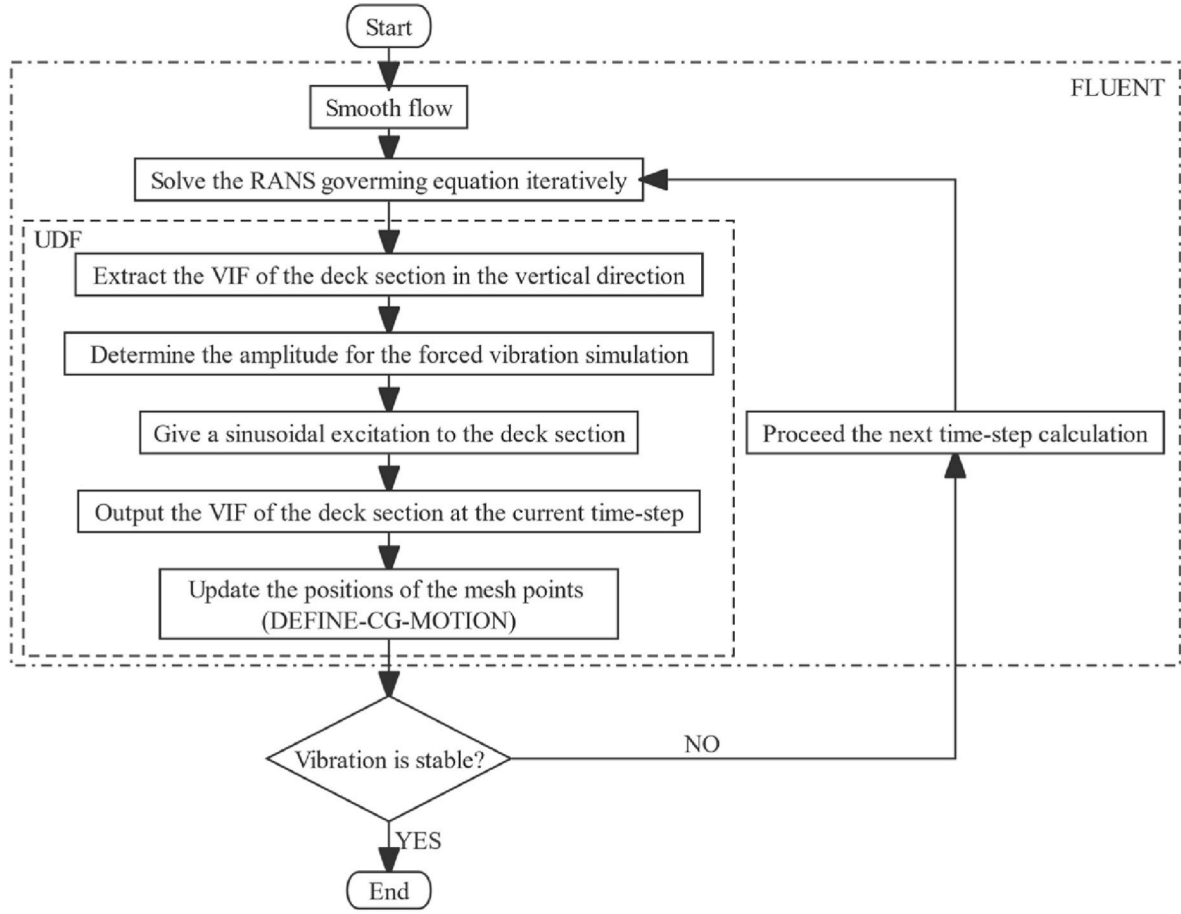


Fig. 7. Flow chart of forced vibration simulation.

time step, which correspond to $T/3200$, $2T/3200$, $3T/3200$, $4T/3200$, $6T/3200$ and $8T/3200$ of the deck section, respectively (T is the natural period of the deck section). For the incoming wind speed of 0.9312 m/s, the maximum displacement response of the deck section is obtained by the afore mentioned aeroelastic vibration simulation. The computed results are compared with the test result obtained by Wang and Chen (2022) and listed in Table 1. In considering both the accuracy and efficiency of the computation, the time step is selected as 0.00201 s in the subsequent simulation.

In this study, the $k-\omega$ SST turbulence model is used to consider the low Reynolds number effects near the wall. Accordingly, the viscous sublayer is modeled as the wall boundary condition and solved to obtain highly accurate results around the walls and ensure that the maximum value of y^+ for each grid scheme is less than 1 during the grid sensitivity analysis. To test the independence of the computation results on the meshing size, six meshing schemes (G1-G6) with different heights (0.0000375 m, 0.00005 m, 0.0000625 m, 0.000075 m, 0.0000875 m and 0.0001 m) of the first layer are generated. The number of grid layers in the boundary and the grid growth rate are adjusted to ensure that the grid size is suitable for the wall boundary condition, as shown in Table 2. The sensitivity study of meshing size is carried out, the maximum y^+ , the number of grids and the computed maximum displacements are listed in Table 3. The G4 scheme is finally selected for the subsequent numerical simulation.

3.4. Computed displacement time history of VIV

According to the sensitivity study in Section 3.3, the time step of 0.00201 s and the grid G4 scheme are selected to carry out the aeroelastic vibration simulation of the deck section. The computed vortex-induced

displacement time history and frequency amplitude spectrum are shown in Fig. 5. It can be seen from the figures that the maximum displacement of the deck section is 3.96 mm when VIV becomes stable. The resonance frequency is 3.087 Hz, which is approximately equal to the natural frequency of 3.104 Hz of the deck section and consistent with the assumption " $\omega_0 \approx \omega_s$ " used in the method of multiple scales. In addition, the displacement time history has obvious periodicity. Except for the main frequency of 3.104 Hz, the vibration amplitudes at other frequencies are very low, which proves that the VIV of the deck section is indeed a weakly nonlinear vibration system.

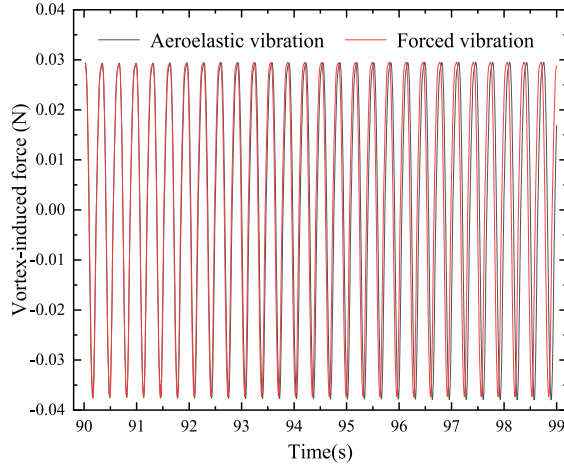
3.5. Identification of aerodynamic parameters

Zhu et al. (2013) proposed an identification method for aerodynamic parameters in the VIF model using the VIF time history measured from the deck section during VIV test and a nonlinear least-squares fitting technique. The rationale behind this method is that the instant work done by the VIF calculated from the VIF model should be equal to the instant work done by the actual VIF acting on the deck section.

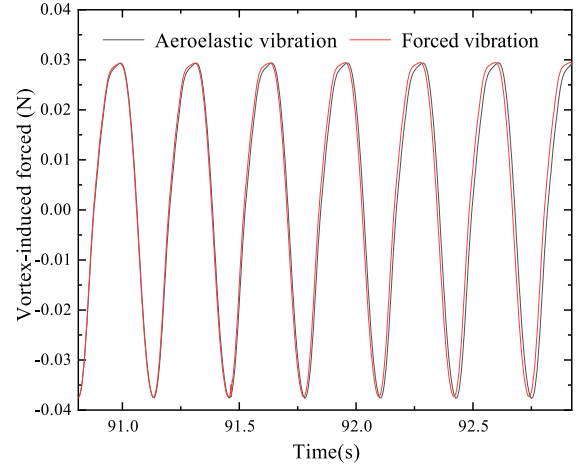
$$\begin{aligned} F_{VI}(x, \dot{x})\dot{x} &= F_{VI}(t)\dot{x} \\ F_{VI}(x, \dot{x})x &= F_{VI}(t)x \end{aligned} \quad (27)$$

where $F_{VI}(x, \dot{x})$ and $F_{VI}(t)$ represents the VIF per unit length calculated from the VIF model and acting on the deck section, respectively.

According to the principle of energy equivalence, the aerodynamic parameters can be identified in an average sense if the work done by the VIF calculated from the VIF model is equal to that by the VIF acting on the deck section within the same vibration period.



(a) 9 seconds



(b) zoom

Fig. 8. Comparison of VIF time histories from aeroelastic and forced vibration simulations.

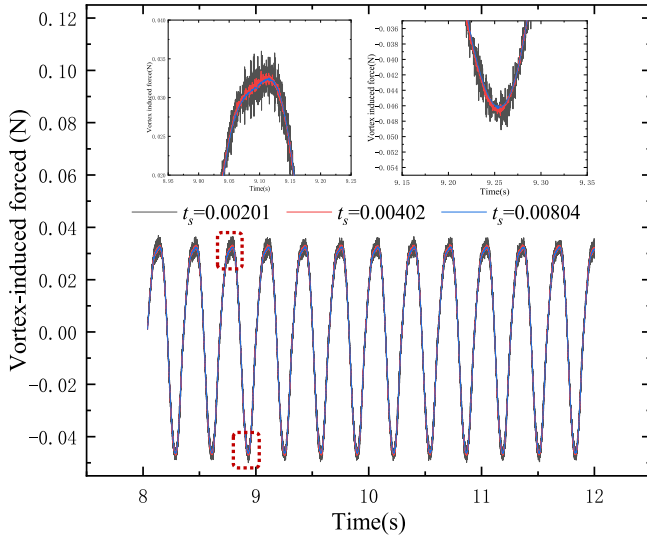


Fig. 9. Sensitivity study of time step.

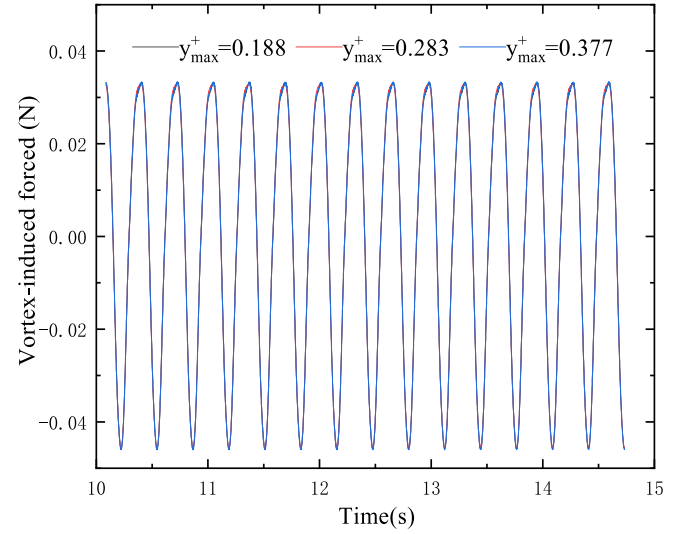


Fig. 10. Sensitivity study of grid size.

$$\begin{aligned} \int_t^{t+nT} F_{VI}(x, \dot{x}) \cdot \dot{x} \, d\tau &= \int_t^{t+nT} F_{VI}(t) \cdot \dot{x} \, d\tau \\ \int_t^{t+nT} F_{VI}(x, \dot{x}) \cdot x \, d\tau &= \int_t^{t+nT} F_{VI}(t) \cdot x \, d\tau \end{aligned} \quad (28)$$

where n is the cycle number in the displacement response time history of the deck section from the initial small amplitude to the stable maximum amplitude of VIV.

In this study, the time history of VIF acting on the deck section is obtained from the aeroelastic vibration simulation of the deck section. The residue (error) function is constructed by the sum of the squares of the error in the instant works done by the VIF calculated from the VIF model and the aeroelastic vibration simulation. The nonlinear least-squares fitting technique is finally used to identify the aerodynamic parameters in the VIF model.

3.6. Verification of the concise VIF model

By using the method of multiple scales with a weakly nonlinear

vibration assumption, the general polynomial VIF model can be reduced to the concise polynomial VIF model, in which the even and cross terms of aerodynamic damping and aerodynamic stiffness in the general polynomial VIF model are neglected because they are much smaller than the odd terms. To verify this point, the aeroelastic vibration simulation of the deck section with a time step of 0.00201s and a G4 grid scheme is performed to obtain the VIV and VIF time histories of the deck section. The first ten aerodynamic damping terms, the first ten aerodynamic stiffness terms, and the first sixteen cross terms in the general polynomial VIF model are identified using the nonlinear least-squares method described in Section 3.5. The two kinds of time histories of VIF are then constructed: one uses the general polynomial VIF model with all the odd, even and cross terms included, and another uses the concise polynomial VIF model with only the odd terms included. The two kinds of VIF time histories are finally compared with the VIF directly from the aeroelastic vibration simulation without any neglect (see Fig. 6).

As can be seen from Fig. 6, the VIF time history given by the concise polynomial VIF model with only the odd terms matches well with the VIF time history given by the general polynomial VIF model in terms of amplitude and period, indicating that the even and cross terms of aerodynamic damping and stiffness have insignificant influence on the

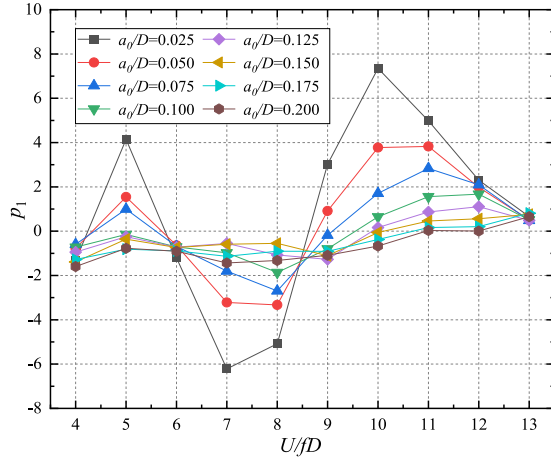
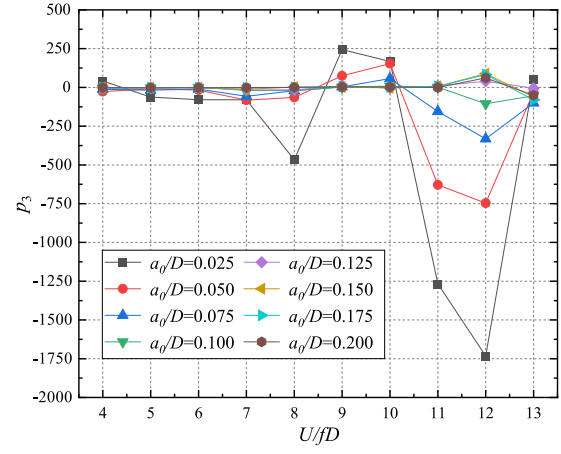
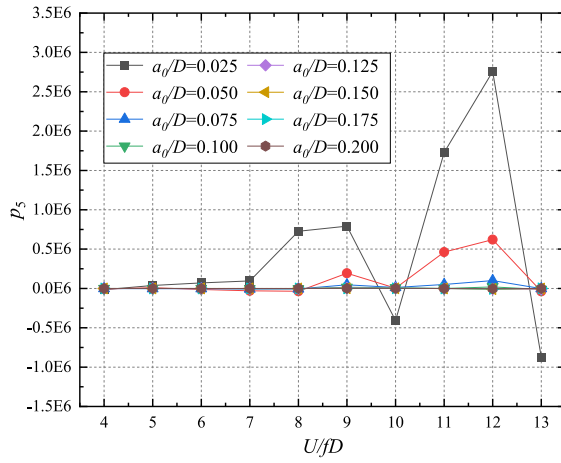
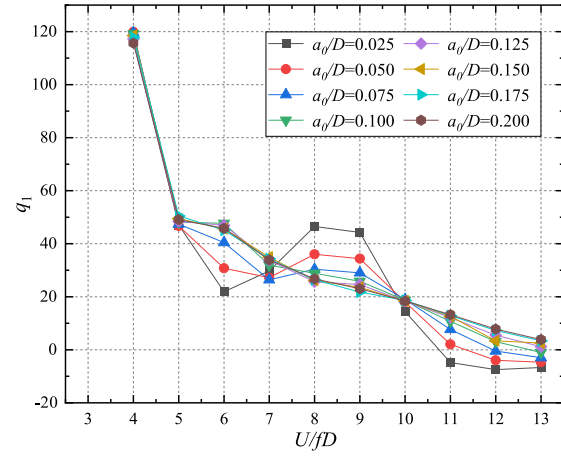
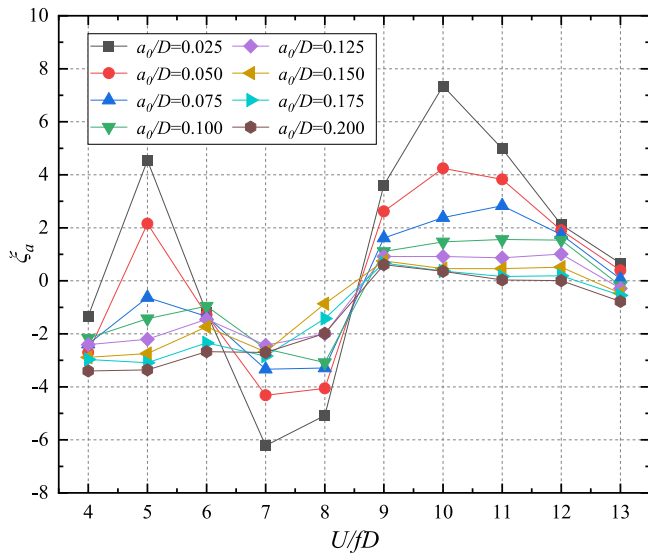
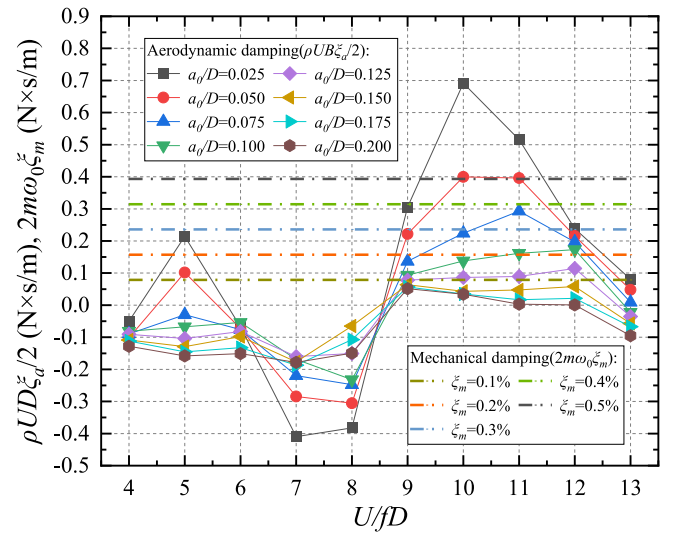
(a) Aerodynamic parameter p_1 (b) Aerodynamic parameter p_3 (c) Aerodynamic parameter p_5 (d) Aerodynamic parameter q_1 **Fig. 11.** Variation of aerodynamic parameters with amplitude and wind speed.**Fig. 12.** Variation of aerodynamic damping ratio ξ_a with wind speed and amplitude.**Fig. 13.** Mechanical damping force and aerodynamic damping forces at different dimensionless wind speeds and amplitudes.

Table 4

Lock-in regions of VIV at different mechanical damping ratios for a dimensionless amplitude of 0.025.

ξ_m	VIV range
0.1%	4.5–5.5, 8.7–13.0
0.2%	4.8–5.2, 8.8–12.5
0.3%	8.9–12.0
0.4%	9.0–11.7
0.5%	9.2–11.4

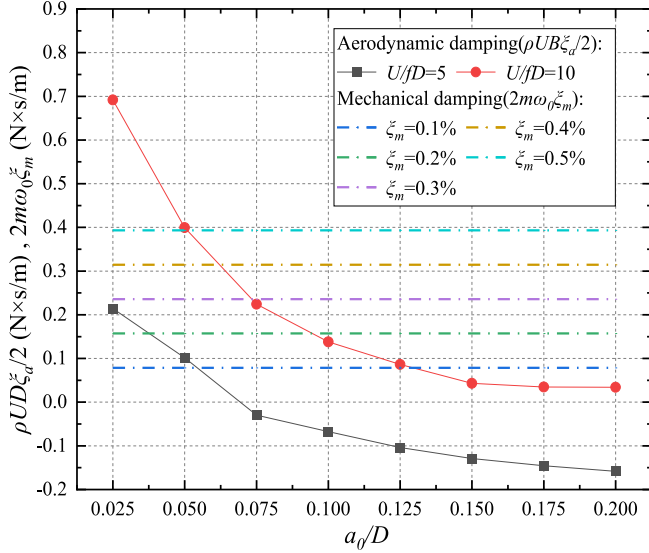


Fig. 14. Mechanical damping force and variation of aerodynamic damping forces with vibration amplitudes.

amplitude and period of VIV. Although there is a slight difference in phase lag (approximately 7.4% of the vibration period) between the two models, it will be shown in Section 5.6 through the comparison between the numerical simulation and experimental results that this small phase lag difference has a negligible impact on the prediction of vortex-induced vibration using the forced vibration method. The assumption of the VIV of the deck section as a weakly nonlinear oscillator and the use of method of multiple scales are rational. Furthermore, the VIV time history given by either the general polynomial VIV model or the concise polynomial VIV model is almost the same as the VIV time history computed from the aeroelastic vibration simulation without any neglect, which indicates that the identification method for aerodynamic parameters has a high accuracy.

4. Simple method for predicting VIR and forced vibration simulation

4.1. Simple method for predicting VIR

According to the research work by Gao and Zhu (2017), the aerodynamic stiffness mainly changes the vibration frequency rather than vibration amplitude of the deck section during VIV. Wind tunnel tests carried out by Diana et al. (2008) also shows that the natural frequency of the deck section varies slightly when VIV occurs. Therefore, the frequency response equation and the critical equation used to predict the vibration amplitude (VIR) can be simplified by neglecting the aerodynamic stiffness terms as

Table 5

Maximum displacement amplitude of VIV at different mechanical damping ratios and reduced wind speeds.

U/fD	5	10
ξ_m		
0.1%	0.0544	0.1295
0.2%	0.0376	0.0940
0.3%	–	0.0734
0.4%	–	0.0621
0.5%	–	0.0509

$$\omega_0^2 a^2 \left(2m\omega_0 \xi_m - \frac{1}{2} \rho U^2 D \sum_{i=1}^n \frac{p_{2i-1} \omega_0^{2i-2} a^{2i-2}}{U^{2i-1}} \right)^2 = \left(\frac{1}{2} \rho U^2 D C_L(K) \right)^2 \quad (29)$$

$$\begin{cases} 2m\omega_0 \xi_m - \rho U^2 D \sum_{i=1}^n \left(\frac{2i-1}{2} \right) \frac{p_{2i-1} \omega_0^{2i-2} a^{2i-2}}{U^{2i-1}} = 0 & (i) \\ \text{or } 2m\omega_0 \xi_m - \frac{1}{2} \rho U^2 D \sum_{i=1}^n \frac{p_{2i-1} \omega_0^{2i-2} a^{2i-2}}{U^{2i-1}} = 0 & (ii) \end{cases} \quad (30)$$

The substitution of Part (ii) of Eq. (30) into Eq. (29), the frequency response equation is reduced to $C_L(K) = 0$, which means no VIV. Thus, only Part (i) of Eq. (30) can be taken as the critical equation.

$$2m\omega_0 \xi_m = \frac{1}{2} \rho U^2 D \sum_{i=1}^n \frac{(2i-1) p_{2i-1} \omega_0^{2i-2} a^{2i-2}}{U^{2i-1}} \quad (31)$$

It is clear now that if the aerodynamic parameters can be identified by using the identification method presented in Section 3.5 for a given wind speed, the amplitude a of VIV of the deck section at this wind speed can be obtained by Eq. (31). Nevertheless, the aeroelastic vibration simulation needs to go through a long “self-excitation” process before the vibration amplitude can be stabilized, as shown in Section 3. The performance of aeroelastic vibration simulation is therefore very time-consuming and the computation cost is very high. To improve the computation efficiency without losing too much accuracy, the forced vibration simulation with a harmonic motion form is used to generate vortex-induced force and response time histories of a bridge deck section and to identify the parameters of aerodynamic damping in Eq. (31).

Moreover, to facilitate the subsequent discussion, the aerodynamic damping ratio ξ_a of VIV is defined as

$$\xi_a = \sum_{i=1}^n \frac{(2i-1) p_{2i-1} \omega_0^{2i-2} a^{2i-2}}{U^{2i-2}} = p_1 + \frac{3\omega_0^2 a^2}{U^2} p_3 + \frac{5\omega_0^4 a^4}{U^4} p_5 + \frac{7\omega_0^6 a^6}{U^6} p_7 + \dots \quad (32)$$

As a result, Eq. (31) can be expressed as

$$2m\omega_0 \xi_m = \frac{1}{2} \rho U D \xi_a \quad (33)$$

The left side of Eq. (33) is the mechanical damping force per unit velocity and the right side of Eq. (33) is the aerodynamic damping force per unit velocity. They are simply called the mechanical damping force and the aerodynamic damping force, respectively, in the subsequent discussion.

4.2. Forced vibration simulation

In the forced vibration simulation, only the fluid solver is used. The deck section is given a sinusoidal excitation with a frequency the same as the natural frequency of the deck section for a certain wind speed U and a given amplitude a . The RANS governing equations for the deck section are iteratively solved using the same fluid solver as used in the aeroelastic vibration simulation. The vortex-induced force and response time histories are obtained without using the structure solver. The procedure of forced vibration simulation is shown in Fig. 7.

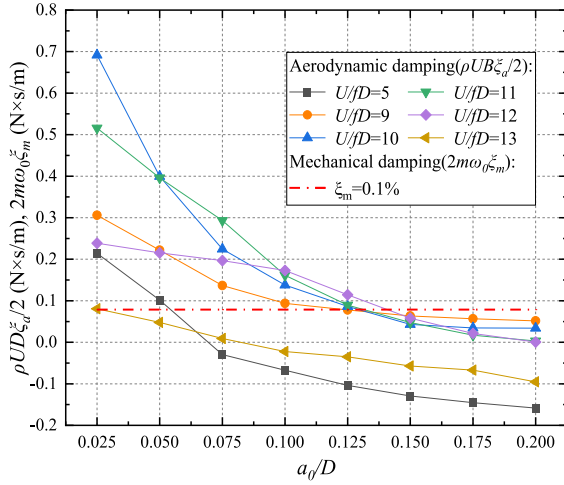
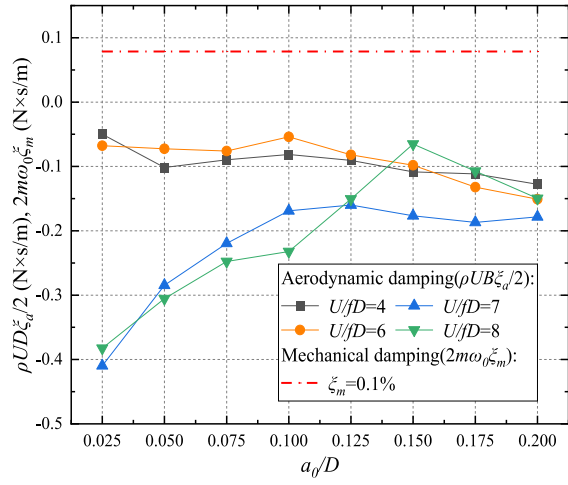
(a) $U/fD=5,9,10,11,12,13$ (b) $U/fD=4,6,7,8$

Fig. 15. Mechanical damping forces and variations of aerodynamic damping forces with dimensionless amplitudes.

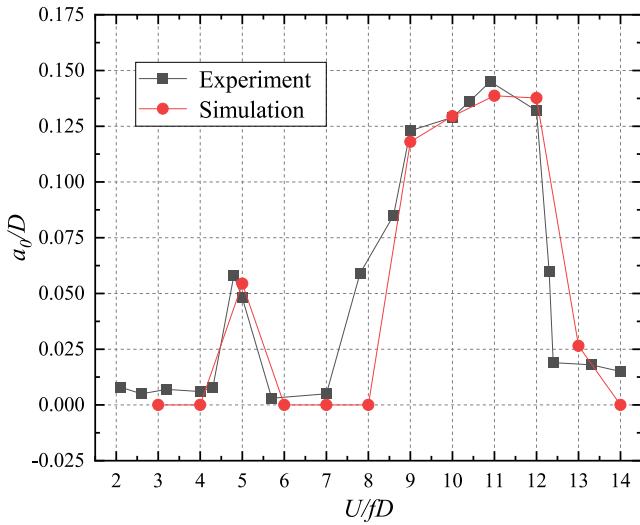


Fig. 16. Variation of dimensionless amplitude with reduced wind speed.

The forced vibration simulation of the deck section is carried out for a series of wind speed U and amplitude a . The vortex-induced force and response time histories is obtained from the forced vibration simulation, and the aerodynamic parameters are then identified. The variations of aerodynamic parameters with wind speed and vibration amplitude can be worked out. The lock-in region and the maximum amplitude of the VIV can be finally obtained by the simple method according to Eq. (31). The case study using the simple method and the forced vibration simulation is given in the following section.

5. Case study

The deck section shown in Fig. 1 is used as a case study. According to Simiu and Scanlan (1986), in the lock-in region of VIV, the harmonic excitation term in the concise polynomial VIF model is very small compared with the odd terms of aerodynamic damping and stiffness. Furthermore, Gao and Zhu (2017) pointed out that the aerodynamic stiffness affects only vibration frequency rather than vibration amplitude. Therefore, this case study uses the concise polynomial VIF model with only the first three odd terms of aerodynamic damping and the first

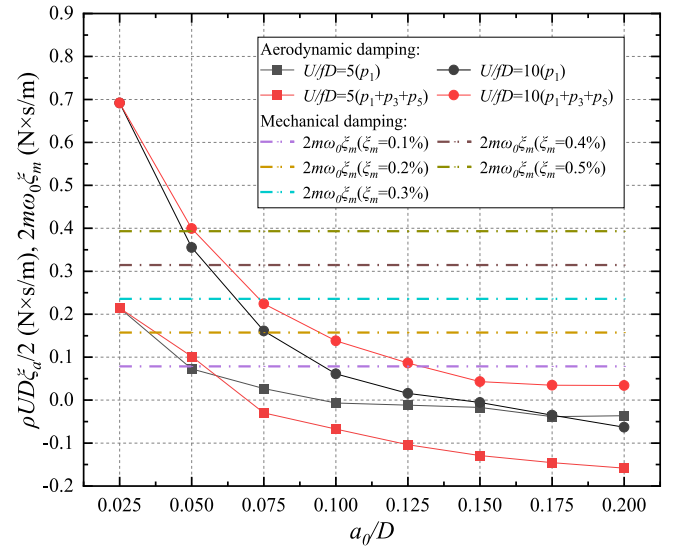


Fig. 17. Variation of mechanical damping forces and aerodynamic damping forces with dimensionless amplitude.

one odd term of aerodynamic stiffness as used in Gao et al. (2020). The equation of motion for VIV of the deck section is given as

$$m\ddot{x} + 2m\omega_0\xi_m\dot{x} + m\omega_0^2x = \frac{1}{2}\rho U^2 D \left(p_1 \frac{\dot{x}}{U} + p_3 \frac{\dot{x}^3}{U^3} + p_5 \frac{\dot{x}^5}{U^5} + q_1 \frac{x}{D} \right) \quad (34)$$

5.1. Verification of forced vibration simulation

As can be seen from Fig. 5, when the wind speed is 0.9312 m/s, the maximum displacement of VIV of the deck section predicted by the aeroelastic vibration simulation is 3.96 mm. The same wind speed and vibration amplitude are used in the forced vibration simulation. The obtained VIF time history from the forced vibration simulation is then compared with that obtained from the aeroelastic vibration simulation, as shown in Fig. 8. It can be found that the period, amplitude and variation of the two VIF time histories are in good agreement, which verify that the forced vibration method can simulate the VIV of the deck section.

To ensure the quality of forced vibration simulation, the sensitivity

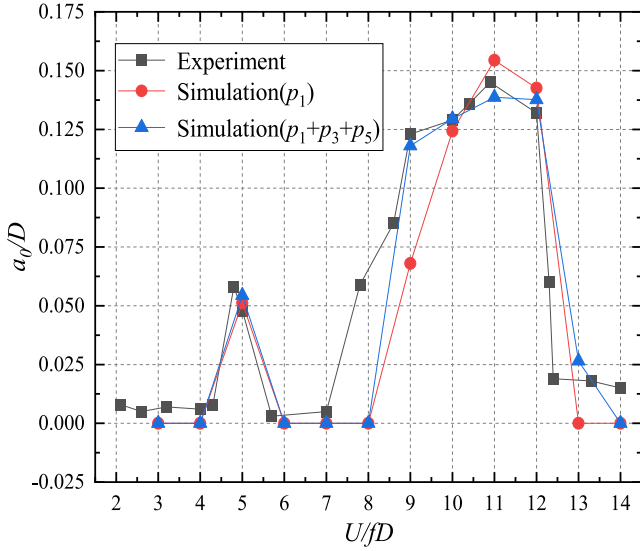


Fig. 18. Experimental and computed results of different aerodynamic damping terms.

study of both time step and mesh size is carried out using three time-steps (0.00201s, 0.00402s, 0.00804s) and three mesh divisions ($y_{\max} = 0.188, 0.283, 0.377$). The corresponding VIV time histories are computed and shown in Fig. 9 and Fig. 10. To balance the computational accuracy and efficiency of the forced vibration simulation, the time step and the grid scheme used in the subsequent numerical simulation are 0.00804s and G2 respectively.

5.2. Aerodynamic parameter identification

With the selected time step and grid size, the forced vibration simulations of the deck section for VIV are performed with the dimensionless amplitude (a_0/D) ranging from 0.025 to 0.200 at an interval of 0.025 and the dimensionless wind speed (U/fD) ranging from 3 to 14 with an interval of 1. The variations of the identified aerodynamic parameters p_1, p_3, p_5 and q_1 with the dimensionless amplitude and wind speed are shown in Fig. 11.

To facilitate the subsequent discussion and according to the general expression of aerodynamic damping ratio (Eq. (32)), the aerodynamic damping ratio with only the first three odd terms can be expressed as

$$\xi_a = p_1 + \frac{3\omega_0^2 a^2}{U^2} p_3 + \frac{5\omega_0^4 a^4}{U^4} p_5 \quad (35)$$

The variation of the aerodynamic damping ratio ξ_a with reduced wind speed and dimensionless amplitude is shown in Fig. 12.

5.3. Prediction of lock-in region of VIV

Let the mechanical damping ratio of the deck section range from 0.1% to 0.5% at an interval of 0.1%. The mechanical damping force and aerodynamic damping force under different dimensionless wind speeds and amplitudes are shown in Fig. 13. Select the horizontal line of mechanical damping ratio of 0.1% and the aerodynamic damping curve of the second smallest dimensionless amplitude of 0.025 (the first smallest amplitude is regarded as zero). The horizontal line of the mechanical damping ratio of 0.1% crosses the aerodynamic damping curve of the dimensionless amplitude of 0.025 at the four points of the reduced wind speed, from which the first and second lock-in regions can be identified as [4.5, 5.5] and [8.7, 13.0] for the level of 0.025 dimensionless amplitude. The prediction results of lock-in regions of VIV of the deck section at different mechanical damping ratios are listed in Table 4 for

the level of 0.025 dimensionless amplitude.

5.4. Prediction of amplitude of VIV at different reduced wind speeds

Let us consider the two reduced wind speeds of 5 and 10. The mechanical damping ratio of the deck section still ranges from 0.1% to 0.5% at an interval of 0.1%. The mechanical damping force and the variation of aerodynamic damping force of the deck section with different vibration amplitudes are shown in Fig. 14.

As can be seen from Fig. 14, when the reduced wind speed is 10 and the mechanical damping ratio is 0.1%, the intersection of the horizontal line of mechanical damping force and the aerodynamic damping force curve gives a vibration amplitude of 0.1295, which is the maximum amplitude of the deck section under the given reduced wind speed and mechanical damping ratio. The prediction results of the maximum amplitude of VIV of the deck section at different mechanical damping ratios are listed in Table 5.

5.5. Comparison with wind tunnel test results

Let us consider that the reduced wind speed ranges from 3 to 14 at an interval of 1 and the mechanical damping ratio of the deck section is 0.1%. The horizontal lines of mechanical damping force and the variations of aerodynamic damping forces of the deck section with different vibration amplitudes are computed and the results are shown in Fig. 15.

Based on the results shown in Fig. 15, considering the lock-in region identified (see Table 3), and using a linear interpolation for zero amplitude, the variation of dimensionless amplitude of the deck with reduced wind speed can be obtained and shown in Fig. 16. Fig. 16 also shows the corresponding wind tunnel test results of the same deck section. The comparison shows that the lock-in regions and the vibration amplitudes of the deck section, predicted by the proposed simple method, are in a good agreement with the wind tunnel test results.

5.6. Effect of higher odd terms of aerodynamic damping force

To investigate the effect of higher odd aerodynamic damping terms on the prediction accuracy of vortex-induced displacement amplitude using the proposed method, the aerodynamic damping forces (per unit velocity) with the first one odd term and first three odd terms are computed and the results are shown in Fig. 17. Furthermore, based on the results shown in Fig. 17, considering the lock-in region identified (see Table 4) and using a linear interpolation for the zero amplitude, the variation of dimensionless amplitude of the deck with reduced wind speed can be obtained and shown in Fig. 18 together with wind tunnel test results. The comparative results show that the prediction accuracy using the first three odd terms is higher than using only the first one odd term.

To have a general understanding of the effect of higher odd aerodynamic damping terms on the prediction accuracy of vortex-induced displacement amplitude using the proposed simple method, let us consider the m -th order odd aerodynamic damping ratio ξ_{am} .

$$\xi_{am} = (2m-1) \frac{p_{2m-1} \omega_0^{2m-2} a^{2m-2}}{U^{2m-2}} \quad (36)$$

According to the discussion in Section 2.2, the vortex shedding frequency is approximately equal to the natural frequency of the deck section when the SDOF system is in the lock-in region of VIV. Therefore, the ratio of the $(m+1)$ -th order odd aerodynamic damping ratio to the m -th order odd aerodynamic damping ratio can be given as

$$\eta = \frac{\xi_{am+1}}{\xi_{am}} = \frac{(2m+1) \cdot \frac{\omega_0^{2m} a^{2m}}{U^{2m}}}{(2m-1) \cdot \frac{\omega_0^{2m-2} a^{2m-2}}{U^{2m-2}}} \approx 4\pi^2 \cdot \frac{2m+1}{2m-1} \cdot St^2 \cdot a_h^2 \quad (37)$$

As can be seen from Eq. (37), this ratio is related to the Strouhal (St) number and dimensionless amplitude of the deck section. The St number

and dimensionless amplitude of the deck section used in this case study are small and equal to 0.1 and 0.145 respectively. The effect of higher odd aerodynamic damping term is thus small. Nevertheless, when the St number and/or dimensionless amplitude of the deck section is large, the higher order aerodynamic damping terms should be considered. The requirement on different numbers of aerodynamic damping terms for different shapes of bridge deck was also pointed out in the literature (Xu et al., 2020, 2023).

6. Conclusions

The method of multiple scales has been used to analyze the general polynomial model for vortex-induced force (VIF) of a bridge deck section. A concise polynomial model for VIF and a simple method for predicting vortex-induced vibration (VIV) of a deck section have been developed. A case study has been also performed to verify the concise VIF model and the simple VIV prediction method. The major conclusions from this study are summarized as follows:

- (1) By assuming that VIV of a deck section as a SDOF is a weakly nonlinear oscillator and using the method of multiple scales, the even and cross terms of the aerodynamic damping and stiffness in the general polynomial VIF model can be neglected and a concise polynomial VIF model with only the odd terms can be derived. Because no specific cross section shape of the bridge deck is assumed in the derivation, the concise polynomial VIF model can be applied broadly.
- (2) The aeroelastic vibration simulation of a deck section is carried out and the aerodynamic parameters in the general and concise polynomial VIF models are identified. The results verify that the even and cross terms of aerodynamic damping and stiffness in the general polynomial VIF model have little influence on the VIF. The assumption of the VIV of the deck section as a weakly nonlinear oscillator and the use of method of multiple scales are rational. The concise polynomial VIF model is accuracy enough.
- (3) The frequency response equation and the critical equation of VIV of a deck section are derived and a simple method for predicting the lock-in region and vibration amplitude of VIV is developed together with the forced vibration simulation of VIV of the deck section.
- (4) The case study of a deck section verifies the accuracy of the forced vibration simulation. The procedure of predicting the lock-in

region and vibration amplitude of VIV of the deck section is demonstrated. The variation of vibration amplitude with reduced wind speed of the deck section, predicted with the simple method, matches well with the wind tunnel VIV test results.

- (5) The effect of higher order aerodynamic damping term on the prediction accuracy of VIV is investigated. The results show that such an effect is related to the St number and dimensionless amplitude of the deck section. If the St number and dimensionless amplitude of the deck section are small, the effect of higher odd aerodynamic damping term is small. When the St number and/or dimensionless amplitude of the deck section is large, the higher order aerodynamic damping terms should be considered.

CRediT authorship contribution statement

Bin Wang: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, preparation, Supervision, Project administration, Funding acquisition. **Shengran Hao:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, preparation, Visualization. **You-Lin Xu:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, preparation. **Yang Liu:** Investigation. **Yongle Li:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix. Nomenclature

Notations	Definition/Explanation
A_0, B_0, C_0, D_0, E_0	The coefficients used in Eq. 19
B	The width of the deck section
$C_L(K)$	The aerodynamic parameter of VIF model
D	The height of the deck section
$D_0, D_1, D_2, \dots, D_n$	Symbols for taking the partial derivatives of $T_0, T_1, T_2, \dots$
F_{VI}	Vortex-induced force (VIF)
$G_1, G_2, G_3, G_4, G_5, G_6$	Grid schemes
K	The reduced frequency of vortex shedding
St	Strouhal number
T	The natural period of the deck section
$T_0, T_1, T_2, \dots, T_n$	Time scales
U	Incoming wind speed
$Y_1(K), Y_2(K)$	Aerodynamic parameters of VIF model
a	The amplitude in the derived solution
a_0	The initial amplitude of forced vibration
i, j, k	Natural number
m	The mass per unit length of the deck section
p_b, q_b, r_{ij}	Aerodynamic parameters of VIF model
t	Time
$x, \dot{x}, \ddot{x}$	The displacement, velocity and acceleration of deck section vibration in vertical direction

(continued on next page)

(continued)

Notations	Definition/Explanation
x_0	Derived solution
y^+	The non-dimensional wall distance for a wall-bounded flow
β	$\beta = \sigma T_1 + \varphi - \theta$
ε	The dimensionless small parameter in multi-scale method
$\varepsilon_{03}, \varepsilon_{04}, \varepsilon_1, \varepsilon_{11}, \varepsilon_{22}$	Aerodynamic parameters of the VIF model
ζ_2	The aerodynamic parameter of the VIF model
η	Ratio of aerodynamic damping of the m+1-th term to the m-th term
θ	The phase difference in the derived solution
ξ_a	Aerodynamic damping ratio
ξ_{am}	The aerodynamic damping ratio of the m-th term
ξ_m	The mechanical damping ratio of the deck section
$o(\varepsilon^n)$	The higher order infinitesimal of ε^n
ρ	Air density
σ	Tuning parameter
φ	The phase difference of the VIF model
ω_0	The natural circle frequency of the deck section
ω_s	The circle frequency of the VIF

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